

TEA FAZIO

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EDUCATION

University of Texas at Austin

B.S. Aerospace Engineering | *Cockrell School of Engineering*

Relevant Courses: Statics, Mechanics of Solids, Dynamics, Thermodynamics, Materials, Composites, Fluids

GPA: 3.91 with University Honors

Expected Graduation: May 2028

PROFESSIONAL EXPERIENCE

Saronic Technologies | Mechanical Integration Team

Austin, TX

Mechanical Engineering Intern

June 2026 – Aug 2026

- Owned mechanical design and integration of 11 component power electronics subsystem for a next-generation autonomous vehicle, driving packaging, mounting, thermal management, and hardware integration, and harnessing from space claim to first article build.
- Designed subsystem mounting hardware and custom heat sinks in Siemens NX, using GD&T drawings for in-house and outsourced production.
- Performed ANSYS FEA on structural components to evaluate stress, deflection, and load paths under 12g multi-axis loads case, using results to optimize for 13% weight reduction from prototype iteration.

Abbott Laboratories | Diabetes Care Research and Development Division

Alameda, CA

Mechanical Engineering Intern

May 2025 – Aug 2025

- Designed universal button life-cycle testing device for glucose monitor remote using a repurposed 3D printer carriage to cycle buttons >100,000 times, eliminating manual testing requirements and unnecessary setup time.
- Designed and built an NFC characterization test fixture with near-zero electromagnetic interference to quantify glucose monitor and remote communication range across 3 axes within 0.1 mm precision.
- Automated test setup by developing Python tooling with Microsoft AI to convert CSV button-coordinate data directly into machine G-code, eliminating manual PC monitoring and saving ~3 work hours per full test setup.

ACTIVITIES

Longhorn Racing | FSAE Electric, Vehicle Dynamics System

Austin, TX

Unsprung (Wheel Assembly) Lead Engineer

Sep 2024 – Present

- Directing project timeline, design, and validation of vehicle wheel assembly as subsystem project manager, leading a team of 4 engineers.
- Collaborating with suspension engineers to establish system-level unsprung compliance targets and design constraints, balancing stiffness, mass, manufacturability, and vehicle performance.
- Redesigned and optimized the rear upright, reducing toe and camber compliance by **26%** while simultaneously reducing wheel assembly mass through iterative CAD and FEA-driven design.
- Developed GD&T drawings for a complex 5-axis CNC-machined rear upright, defining functional datums, fits, and geometric controls.
- Led manufacturing of the unsprung system, producing 8 unique CNC-machined components in-house while coordinating additional outsourced parts, inspection, and assembly.
- Designed, manufactured, and validated steering gearbox system featuring split housing with custom bevel gears, incorporating preloaded bearing assemblies with retention nuts, reducing system weight by 15% while maintaining stiffness.

Longhorn Racing | Executive Officer Board

Austin, TX

Outreach Officer

Dec 2025 – Present

- Led recruitment for 350 members, designing and executing an application cycle that generated 800+ unique applicants and 2,000+ applications, the largest applicant pool in organization history.
- Led development and deployment of a custom high-traffic recruiting platform, partnering with software engineers to define requirements, debug workflows, and improve reliability; supported 300+ reviewers managing thousands of applications and candidate evaluations.
- Coordinated recruiting operations across three leadership teams, including reviewer training, candidate communications, interview and trial logistics, informational sessions, and selection timelines to ensure a consistent process across the organization.
- Owned recruiting marketing and applicant outreach, coordinating advertising, campus promotion, and informational events to expand Longhorn Racing's reach and drive record applicant engagement.

SKILLS

Siemens NX, SOLIDWORKS, ANSYS FEA, Fusion360 CAM, CNC Lathe/Mill, GD&T, Rapid Prototyping, Project Management, Electrical Integration/Enclosures